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Subject: PL-II Roll No.: 63 Exp. No.: 2
Name of the Experiment:

Performed on:
Submitted on:

Marks	Teacher's Signature with date
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PL Assignment - 2

1. list the python sequence data types and state one key characterics of each with respect to mutability or ordering.

Data types are :

- String (str) : immutable and ordered collection of characterics.
- list : Mutable & ordered collection of elements
- Tuple : immutable and ordered collection of elements
- Range : Immutable and ordered sequence of numbers.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

2. Explain how indexing and slicing work in python sequences. Illustrate how they differ when applied to strings & lists

Indexing is used to access a single element of a sequence using its position. Positive indexing starts from -1.

slicing extracts a part of sequence using the syntax.

Difference in strings & lists:

- In strings, indexing / slicing returns characterics or substring.

- In lists, indexing / slicing returns elements or sublists.

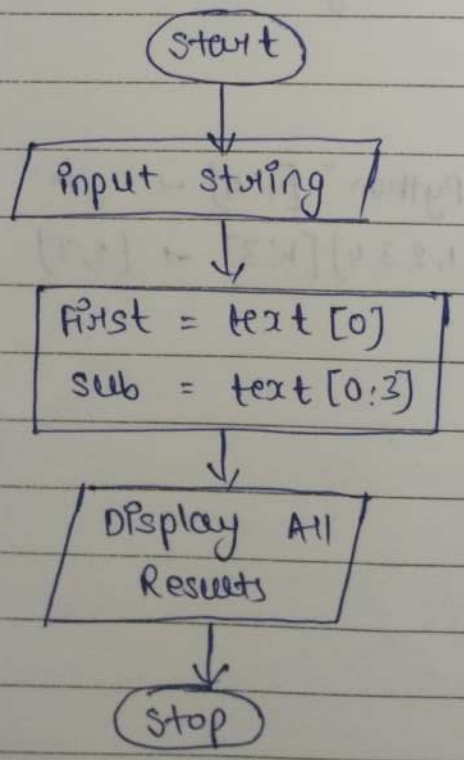
Example: String :- "Python" [1:4] → "yth"
list : [1,2,3,4][1:3] → [2,3]

3. python program using indexing, slicing & build in functions.

code :-

```
text = "Engineering"
first - char = text [0]
sub - text = text [0:5]
Length = len (text)
upper_text = text.upper ()

print ("First character :", first - char)
print ("sliced text :", sub - text)
print ("length :", length)
print ("uppercase :", upper text)
```



4. Difference between equality (==) and identity is

- == (Equality operator) checks whether 2 sequences have the same values.

- is (identity operator) checks whether 2 variables refer to the same object in memory.

eg. a = [1,2]
b = [1,2] → a is b is True, but a is b is False

5. Evaluation of lists, sets & dictionaries for student records.

- lists allows duplicates and are ordered, but searching is slower.

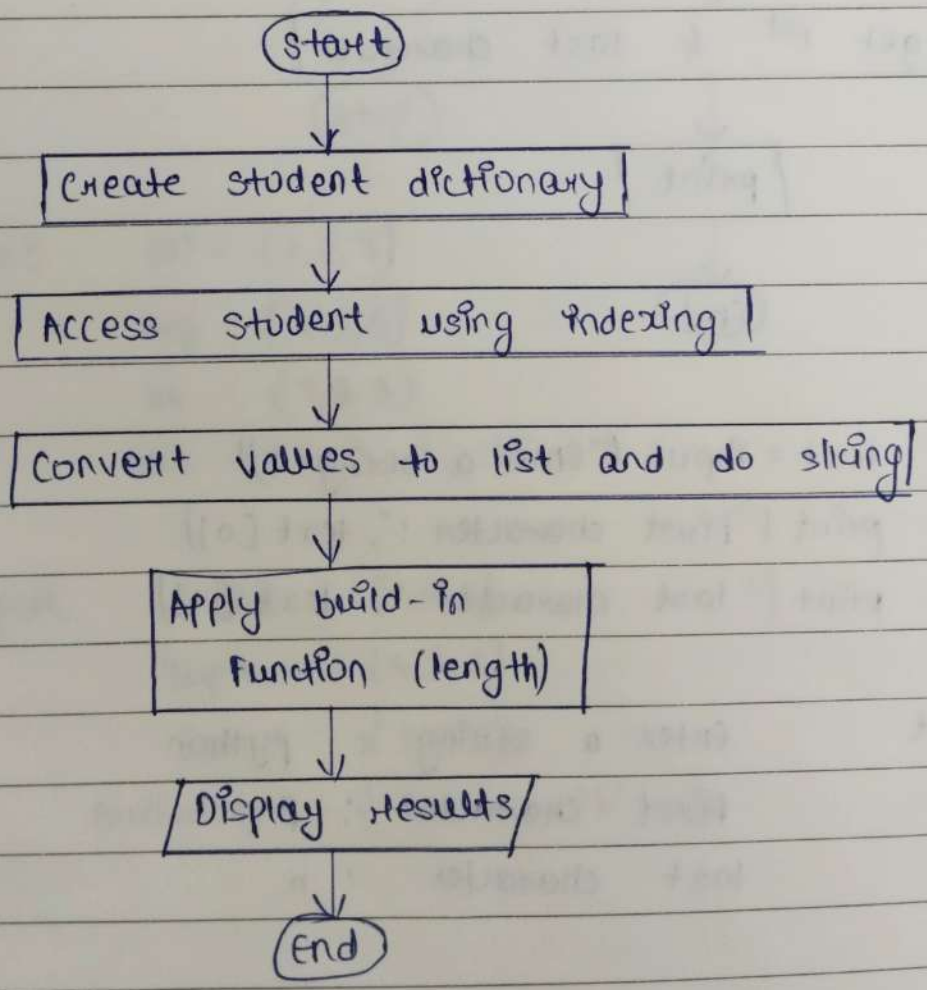
- Sets store only unique values and are fast for membership testing.

- Dictionaries store data in key-value pairs and allow fast access using keys

Most effecient : Dictionaries are best for student record because each student can be stored with a unique ID as a key, enabling fast lookup, update & retrieve.

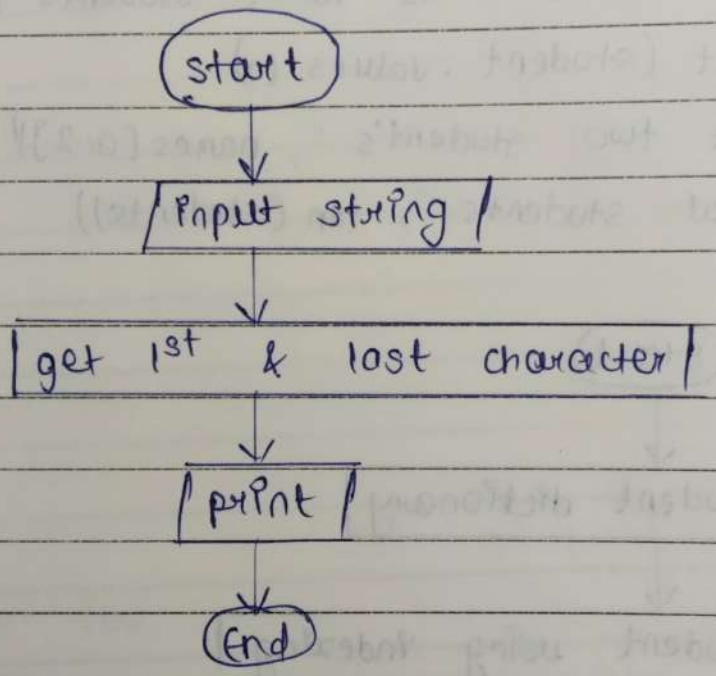
6. Program to manage student data

```
students = {
    101 : "Aman",
    102 : "Riya",
    103 : "Karan".
}
print ("Student with ID 101 :" students [101])
names = list (student .values ())
print ("First two student's ", names [0:2])
print ("Total students:", len (students))
```



1. Print first & last character of a string.

Algorithm : 1. Start
2. Input a string
3. Print 1st character using indexing - 0.
4. Print last character using index - 1
5. End



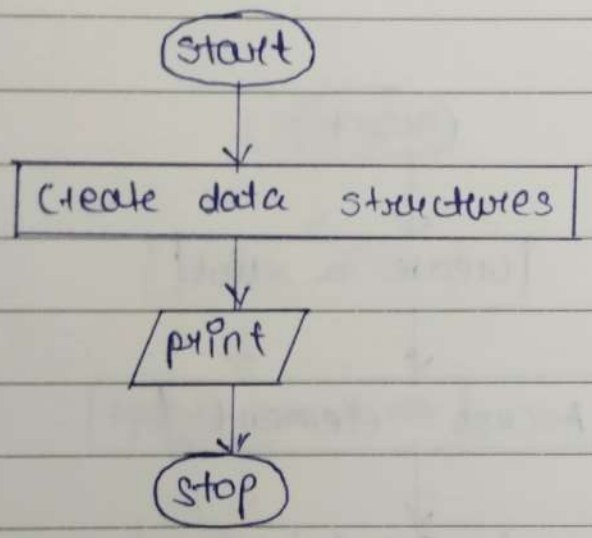
Code :
```
text = input ("Enter a string : ")
print ("First character :", text [0])
print ("last character :", text [-1])
```

Output
```
Enter a string : python
First character : p
last character : n
```

8. Create list, tuple, set, & dictionary.

Algorithm,
1. Start
2. Create list, tuple, set, dictionary
3. Display them
4. Stop

Flowchart,



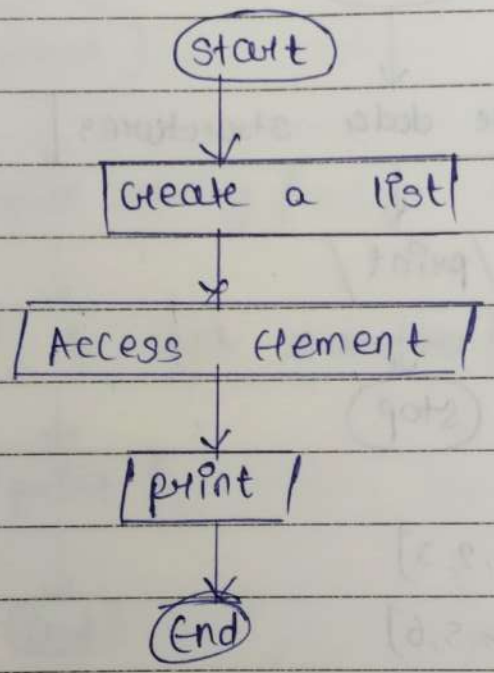
Code:
```
lst = [1,2,3]
tup = [4,5,6]
st = {7,8,9}
dct = {"a": 1, "b": 2}
```

Output,
```
List : [1,2,3]
Tuple : (4,5,6)
Set : {7,8,9}
Dictionary: {"a": 1, 'b': 2}
```

9. Program to understand Indexing.

Algorithm:
1. Start
2. Create a list
3. Access elements using Indexing
4. Print elements
5. End

Flowchart:



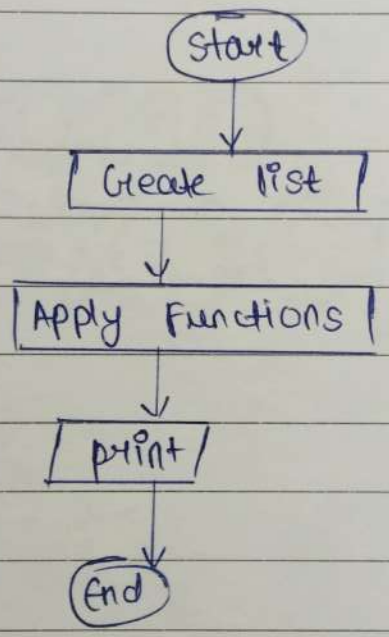
Code:
```
numbers = [10,20,30,40]
print (" First element :" numbers [0]))
print ( "Second element :" numbers [1])
print( " last element :" , numbers [-1])
```

Output:
```
First element : 10
Second element : 20
last element : 40
```

10. program to understand build - in functions

Algorithm :
1. Start
2. Create a list
3. Apply build - in functions
4. print results
5. End

Flowchart :



Code :
```
numbers = [5,10,15,20]
print ("Length :", len(numbers))
print ("Maximum:", Max (numbers))
print ("Minimum :", Min (numbers))
print ("sum :", sum (numbers))
```

Output
```
Length : 4
Max : 20
Min : 5
Sum : 50
```